DIAPO

DESCRIPTION

DIAPO Tablet 5mg
A 8mm diameter, round, flat, scored yellow tablet.
Each tablet contains Diazepam 5mg

INDICATIONS

Diazepam is a long-acting benzodiazepine with anticonvulsant, anxiolytic, sedative and muscle relaxant properties. It is indicated as an antianxiety agent, as a hypnotic in the short-term management of insomnia, as a sedative and preoperative medication, as an anticonvulsant particularly in the management of status epilepticus and febrile convulsions, in the control of muscle spasm as in tetanus, and in the management of alcohol withdrawal symptoms. It is also used as antipanic and antitremor agent. Diazepam is also used to relieve spasms of facial muscles associated with problems of occlusion and temporomandibular joint disorders and for the treatment of tension headache.

PHARMACODYNAMICS

In general, benzodiazepines act as depressants of the central nervous system producing all level of CNS depression from mild sedation to hypnosis to coma depending on the dosage. The precise sites and mechanisms of action have not been completely established. Although various mechanisms of action have been proposed, it is believed that diazepam enhance or facilitate the inhibitory neurotransmitter action of gamma-aminobutyric acid (GABA), which is one of the major inhibitory neurotransmitters in the brain and mediates both pre- and post-synaptic inhibition in all regions of the CNS, following interaction between the benzodiazepine and a specific neuronal membrane receptor. Diazepam reportedly act as agonists at the benzodiazepine receptors, which have been shown to form a component of a functional supramolecular unit known as the benzodiazepine-GABA receptor-chloride ionophore complex. The receptor complex, believed to reside on neuronal membranes that regulate cell firing, functions mainly in the gating of the chloride channel. Activaton of the GABA receptor results in the opening of the chloride channel, allowing the flow of chloride ions through the neuronal membrane. Usually this results in hyperpolarisation of the post-synaptic neuron, which inhibits firing of that neuron. That inhibition translates into decreased neuronal excitability, thus attenuating subsequent depolarisation excitatory transmitters.Benzodiazepines also appear to act at GABA-independent receptors. Its sedative-hypnotic effects are believed to be produced by stimulating GABA receptors in the ascending reticular activating system. Since GABA is inhibitory, receptor stimulation increases inhibition and blocks both cortical and limbic arousal following stimulation of the brain stem reticular formation.

It appears to act as anticonvulsant, at least partially, by enhancing pre-synaptic inhibition. It suppresses the spread of seizure activity produced by epileptogenic foci in the cortex, thalamus, and limbic structures but does not abolish the abnormal discharge of the focus. The exact mechanism of action of diazepam as skeletal muscle relaxant adjunct has not completely established but it appears to produce skeletal muscle relaxation primarily by inhibiting spinal polysynaptic afferent pathways; however, monosynaptic afferent pathways may also be inhibited. It may also directly depress motor nerve and muscle function.

PHARMACOKINETICS

Diazepam is readily and completely absorbed from the gastointestinal tract with, peak plasma concentrations occuring within about 0.5 to 2 hours of oral administration. It is highly lipid soluble and crosses the blood-brain barrier. It has a biphasic half-life with an initial rapid distribution phase followed by a prolonged terminal elimination phase of 1 or 2 days. Its action is further prolonged by the even longer half-life of 2 to 5 days of its principal active metabolite, desmethyldiazepam (nordazepam). Diazepam and desmethyldiazepam accumulate on repeated administration and the relative proportion of desmethyldiazepam in the body increases on long-term administration. No simple correlation has been found between plasma concentrations of diazepam or its metabolites and their therapeutic effect. Diazepam is extensively metabolised in the liver and in addition to desmethyldiazepam, its active metabolites include oxazepam, and temazepam. It is excreted in the urine, mainly in the form of its metabolites, either free or in conjugated form. Diazepam is extensively bound to plasma proteins (98 to 99%). The plasma half-life of diazepam and/or its metabolites is prolonged in neonates, in the elderly, and in patients with liver disease. In addition to crossing the blood-brain barrier, diazepam and its metabolites also cross the placental barrier and are excreted in breast milk.

DOSAGE

Antianxiety
Oral, 2 to 10 mg two to four times a day.
Sedative-hypnotic -Alcohol withdrawal
Oral, 10 mg three to four times a day during the first twenty-four hours, the dosage decrease to 5 mg three or four times a day as needed.
Anticonvulsant
Oral, 2 to 10mg two to four times a day.
Skeletal muscle relaxant adjunct
Oral, 2 to 10mg three to four times a day.
Note:
Debilatated patients
Debilatated patients -Oral, 2 to 2.5 mg one or two times a day, the dosage being increased gradually as needed and tolerated.
Geriatric
Antianxiety, sedative-hypnotic, anticonvulsant or skeletal muscle relaxant adjunct.
Oral, 2 to 2.5 mg one or two times a day, the dosage being increased gradually as needed and tolerated.

SIDE EFFECTS

Drowsiness, sedation, and ataxia are the most frequent adverse effects of diazepam use. They generally decrease on continued administration and are consequences of CNS depression. Less frequent effects include vertigo, headache, confusion, mental depression, slurred speech or dysarthria, changes in libido, tremor, visual disturbances, urinary retention or incontinence, gastrointestinal disturbances, changes in salivation, and amnesia. Some patients may experience paradoxical excitation which may lead to hostility, aggression, and disinhibition. Jaundice, blood disorders, and hypersensitivity reactions have been reported rarely. Respiratory depression and hypotension occasionally occur with high dosage. Rebound anxiety and insomnia may be a result of tolerance to the effects of diazepam or part of a withdrawal syndrome.

Over dosage can produce CNS depression and coma or paradoxical excitation. However, fatalities are rare when taken alone. Use of diazepam in the first trimester of pregnancy has been associated with various congenital malformations in the infant but no clear relationship has been established. Administration of diazepam in late pregnancy has been associated with intoxication of the neonate.

CONTRAINDICATIONS

Acute or predisposition to glaucoma, angle-closure, myasthenia gravis, severe chronic obstructive pulmonary disease, acute alcohol intoxication with depressed vital signs, coma or shock, history of drug abuse or dependence, history of epilepsy or seizures, hepatic function impairment, hyperkinesis, hypoalbuminemia, severe mental depression, organic brain disorders, porphyria, psychoses, renal function impairment, sensitivity to benzodiazepines, established or suspected sleep apnea or swallowing abnormality in children are the risk -benefits that should be considered.

PRECAUTIONS

Diazepam should be avoided in patients with pre-existing CNS depression or coma, acute pulmonary insufficiency, or sleep apnoea, and used with care in those with chronic pulmonary insufficiency. Diazepam should be given with care to the elderly or debilitated patients who may be more prone to adverse effects. Caution is required in patients with muscle weakness, or impaired liver or kidney function. The sedative effects of diazepam are most marked during the first few days of administration; affected patients should not drive or operate machinery. Sedation or respiratory and cardiovascular depression may be enhanced by other drugs with CNS-depressant properties; these include alcohol, antidepressants, antihistamines, antipsychotics, general anaesthetics, other hypnotics or sedatives, and opioid analgesics. Monitoring of cardiorespiratory function is generally recommended when benzodiazepines are used for conscious sedation. Diazepam should not be used for the treatment of chronic psychosis or for phobic or obsessional states.

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even in therapeutic doses for short periods. Because of its dependence liability, diazepam should be used with caution in patients with a history of alcohol or drug addiction.

Laboratory test values
Benzodiazepines may decrease thyroidal uptake of I-123 and I-131.

Risks from Concomitant Use with Opioids
Profound sedation, respiratory depression, coma, and death may result from the concomitant use of Diapo Tablet with opioids. Observational studies have demonstrated that concomitant use of opioids and benzodiazepines increase the risk of drug-related mortality compared to use of opioids alone. Because of these risks, reserve concomitant prescribing of these drugs for use ·in patients for whom alternative treatment options are inadequate.

If the decision is made to newly prescribe a benzodiazepine and an opioid together, prescribe the lowest effective dosages and minimum durations of concomitant use.
If the decision is made to prescribe a benzodiazepine in a patient already receiving an opioid, prescribe a lower ·initial dose of the benzodiazepine than indicated in the absence of an opioid, and titrate based on clinical response.
If the decision is made to prescribe an opioid in a patient already taking a benzodiazepine, prescribe a lower initial dose of the opioid, and titrate based on clinical response.

Follow patients closely for signs and symptoms of respiratory depression and sedation. Advise both patients and caregivers about the risks of respiratory depression and sedation when Diapo Tablet is used with opioids. Advise patients not to drive or operate heavy machinery until the effects of concomitant use of the opioid have been determined. Screen patients for risk of substance use disorders, including opioid abuse and misuse, and warn them of the risk for overdose and death associated with the use of opioids (See Drug Interactions).

Warning
Anaphylaxis (severe allergic reaction) and Angioedema (severe facial swelling) can occur as early as the first time the product is taken. Complex sleep-related behaviors which include sleep driving, making phone calls, preparing and eating food (while asleep) may also occur.

Effects on ability to drive and use machines:
This medicine can impair cognitive function and can affect a patient's ability to drive safely.Patients should be warned that effects on the central nervous system may persist into the day after administration even after a single dose.

OVERDOSAGE

Continuing confusion, decreased reflexes, severe drowsiness, shakiness, slow heartbeat, shortness of breath, or troubled breathing, continuing slurred speech, staggering and severe weakness are all symptoms of diazepam overdose.

For benzodiazepine intoxication, treatment consists of general supportive therapy and includes the following:

a) If the patient is conscious (and not at risk of becoming obtunded, comatose, or convulsing based on ingestion), emesis should be induced mechanically or with emetics. Also activated charcoal may be administered orally to increase clearance as well as decrease absorption of the benzodiazepine.
b) If the patient is unconscious, gastric lavage may be performed with a cuffed endotracheal tube in place to prevent aspiration of vomitus.
c) Respiration, pulse, and blood pressure should be monitored.
d) Oxygen should be administered if respiration is depressed.
e) Intravenous fluids may be administered to promote diuresis.
f) An adequate airway should be maintained.
g) Hypotension may be controlled, if necessary, by intravenous administration of vasopressors such as dopamine, norepinephrine, or metaraminol.
h) If excitation occurs, barbiturates should not be used since they may exacerbate excitation and/or prolong CNS depression.
i) Dialysis is of limited value in the treatment of overdose.

DRUG INTERACTIONS

Concurrent use of alcohol or CNS depressant-producing medications with diazepam may increase the CNS depressant effects of either these medications of diazepam. Caution is recommended and dosage of one or both agents should be reduced when a benzodiazepine is used concurrently with an opioid analgesic, the dosage of the opioid analgesic should reduced by at least one-third and administered in small increments. Precaution is recommended for prolonged concurrent use of addictive medications, especially CNS depressants with habituating potential with diazepam, which may increase the risk of habituation. Concurent use of antacids with diazepam may delay, but not reduce, the absorption of diazepam. Concurrent use of carbamazepine with benzodiazepine metabolized via hepatic enzyme system may result in increased metabolism, leading to decreased serum concentration and reduced elimination half-live of benzodiazepines because of induction of hepatic microsomal enzyme activity; monitoring of carbamazepine blood concentrations as a guide to dosage is recommended, especially when carbamazepine is added to or withdrawn from existing benzodiazepine therapy. Concurrent use of cimetidine, oral estrogen-containing contraceptives, disulfiram, erythromycin or isoniazid with diazepam may inhibit the hepatic metabolism of diazepam, resulting in delayed elimination and increased plasma concentrations. In contrast, concurrent use of rifampicin with diazepam may enhance the elimination of diazepam, resulting in decreased plasma concentrations. Collapse, sometimes accompanied by respiratory depression or arrest, has been reported in a few patients receiving clozapine concurrently with benzodiazepines. Caution is required and the dosage of clozapine should be titrated upwards slowly. Pre-medication with diazepam may decrease the dose of a fentanyl derivative required for induction of anaesthesia and decrease the time to loss of consciousness with induction doses. Concurrent use of hypotension-producing medications with benzodiazepine pre-anaesthetics used in surgery may potentiate the hypotensive response and dosage adjustments may be necessary. Concurrent use of levodopa with benzodiazepines may decrease the therapeutic effects of levodopa. The elimination of diazepam may be prolonged with concurrent use of omeprazole. Concurrent use of zidovudine with benzodiazepines may, in theory, competitively inhibit hepatic glucuronidation and decrease the clearance of zidovudine; the toxicity of zidovudine potentially could be increased.

Opioids
Due to additive pharmacologic effect, the concomitant use of opioids with benzodiazepines increase the risk of respiratory depression, profound sedation, coma and death. The concomitant use of opioids and benzodiazepines increases the risk of respiratory depression because of actions at different receptor sites in the central nervous system that control respiration. Opioids interact primarily at μ-receptors, and benzodiazepines interact at GABAA sites. When opioids and benzodiazepines are combined, the potential for benzodiazepines to significantly worsen opioid-related respiratory depression exists.

Reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate (see Warnings and Precautions).
Limit dosage and duration of concomitant use of benzodiazepines and opioids, and follow patients closely for respiratory depression and sedation.

PREGNANCY AND LACTATION

The hazards of administration of diazepam during pregnancy should be noted.
Benzodiazepine can cause fetal harm when administered to pregnant women. Their use during the first trimester of pregnancy should almost always be avoided.
Patients should be advised that if they become pregnant or intend to become pregnant during benzodiazepine therapy, they should communicate with their doctor about the desirability of discontinuing the drug.
Since many benzodiazepines are distributed into milk and because of the potential for adverse actions reactions from the drugs in nursing infants, doctor advise should be seek taking into account the importance of the drug to the women.

PRESENTATION

Blister pack of 100x10's

STORAGE CONDITIONS AND USER INSTRUCTIONS

Store in a dry place below 30°C.
Keep container tightly closed.
Protect from light.
Keep out of reach of children.
Jauhi daripada kanak-kanak.
Shelf life: Please refer to outer package.

Route of administration: Oral

Product Registration Holder :
Duopharma Manufacturing (Bangi) Sdn. Bhd.
Lot No. 2, 4, 6, 8 & 10, Jalan P/7, Section 13,
Bangi Industrial Estate, 43650
Bandar Baru Bangi, Selangor, Malaysia.

Manufacturer :
Duopharma Manufacturing (Bangi) Sdn. Bhd.
Lot No. 2 & 4, Jalan P/7, Section 13, Bangi Industrial Estate,
43650 Bandar Baru Bangi, Selangor, Malaysia.

DUOPHARMA

1500008655 N.1
Revision Date: Aug 2019

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